



GATR: a graph attention deep reinforcement learning approach with variance-sensitive rewards for rpl routing optimization

Ying Wang^{1,2} · Jianjun Lei¹ · Yuanyuan Li³ · Fengjun Shang¹

Received: 19 May 2025 / Accepted: 1 September 2025 / Published online: 23 October 2025
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Abstract

The Internet of Things (IoT) has become the core infrastructure supporting intelligent perception and interconnection. However, the dynamic network topology, limited resources, and partial observability pose significant challenges to traditional IoT routing algorithms in balancing energy efficiency, reliability, and stability optimally. Therefore, this paper proposes a graph attention deep reinforcement learning-based routing optimization algorithm (GATR), which can optimize the data transmission performance through multi-agent collaborative decision-making. GATR integrates the Actor-Critic architecture with centralized training and decentralized execution (CTDE), and innovatively introduces a topology-aware dual-channel graph modeling mechanism to dynamically aggregate neighborhood information while explicitly encoding physical interference relationships, thereby mitigating partial observability constraints. For resource-constrained terminal devices, GATR designs a lightweight three-layer convolutional Actor network architecture, and introduces contrastive learning to enhance the feature representation of the routing decision model. Furthermore, a variance-sensitive multi-objective reward function formulation is proposed to dynamically balance energy consumption, reliability, and stability through adaptive weight adjustment. Simulation results demonstrate that GATR achieves significant improvements in energy efficiency, network lifetime, and convergence performance compared with RARL, DDQN, DADR, CARL, and MRHOF algorithms, exhibiting excellent environmental adaptability and engineering application potential.

Keywords Internet of things · Routing optimization · Graph attention · Deep reinforcement learning · Topology-aware

Mathematics Subject Classification (2010) Primary35A01 · 65L10 · 65L12 · 65L20 · 65L70

1 Introduction

The rapid development of Internet of Things (IoT) technology has enabled its widespread adoption in various fields including smart cities (Kim et al. 2023), industrial automation (Barbosa et al. 2025), smart healthcare (Li and Huang 2023), and environmental monitoring (Abatal et al. 2024), facilitating massive device connectivity and data exchange. According to industry forecasts, the global number of IoT devices is expected to surpass 29 billion by 2030, with networks exhibiting three prominent characteristics: large scale, significant heterogeneity, and highly dynamic topologies (Chen et al. 2025). Meanwhile, it has made the establishment of efficient and reliable routing mechanisms a critical challenge in IoT system design (Lalani et al. 2024). IPv6 routing protocol for low-power and lossy networks (RPL), as the standard routing protocol for resource-constrained wireless IoT environments, is initially designed for relatively

✉ Jianjun Lei
Leijj@cqupt.edu.cn

Ying Wang
ywang@cqupt.edu.cn

Yuanyuan Li
Liyy@cqupt.edu.cn

Fengjun Shang
Shangfj@cqupt.edu.cn

¹ School of Computer Science and Technology, Chongqing University of Posts and Telecommunications, 400065 Chongqing, China

² School of Artificial Intelligence, Chongqing University of Posts and Telecommunications, 400065 Chongqing, China

³ School of Automation, Chongqing University of Posts and Telecommunications, 400065 Chongqing, China

static network conditions (Zilberman et al. 2024). Hence, it faces significant challenges in highly dynamic and dense deployment scenarios, particularly in terms of energy efficiency, topology adaptability, and quality of service (QoS) assurance. Dynamic network environment and rapid energy consumption often lead to delayed routing updates, resulting in unstable communication paths, higher packet loss rates, and reduced network lifetime. To address these limitations, researchers have proposed various enhanced routing metrics such as expected transmission count (ETX), residual energy, link quality, hop count, and delay sensitivity (Darabkh et al. 2022), along with many RPL-based optimization strategies employing composite objective functions. These approaches demonstrate improved routing accuracy and energy efficiency in static or low-dynamic environments. However, they suffer from several fundamental limitations (Changazi et al. 2024). Specifically, most methods predominantly focus on single-objective optimization (e.g., shortest path or minimal energy consumption), lacking comprehensive multi-objective coordination among critical QoS parameters, including throughput, load balancing, and delay sensitivity (Changazi et al. 2024). Moreover, these strategies typically depend on static parameter configurations and heuristic rules, rendering them inadequate for real-time adaptation to dynamic topology changes, channel variations, and node state fluctuations (Fedila et al. 2025). Furthermore, when confronted with the inherent high-dimensional state space and non-stationary environmental dynamics of IoT systems, conventional methods exhibit constrained policy generalization capabilities, decision-making efficiency, and system scalability (Wang et al. 2024). Although the heuristic-based RPL optimization algorithm has certain practical value, it can not meet the requirements of an efficient, robust, and adaptive routing mechanism for complex, large-scale IoT deployments.

Deep reinforcement learning (DRL) has emerged as a promising alternative, offering end-to-end modeling capabilities and online policy optimization for IoT routing challenges (Khansari et al. 2024). Unlike traditional heuristic approaches, DRL autonomously learns state-action-reward mappings through environmental interactions without requiring explicit network evolution models, demonstrating superior adaptability and generalization, particularly in high-dynamics, complex state-space environments (Wang et al. 2023; Sanchez et al. 2025). Recently, many studies have applied Q-learning, Deep Q-network (DQN), and Double DQN (DDQN) to optimize the RPL protocol (Saqhib and Lakshmikanth 2025; Das 2025; Hou et al. 2023; Abdmeziem et al. 2024). However, most existing work focuses solely on local node states while neglecting the crucial influence of topological structures and inter-node correlations on routing behavior. This oversight leads to suboptimal performance in scenarios featuring rapid connectivity changes or partial

observability constraints. To enhance DRL's environmental modeling capacity, researchers have turned to graph neural networks (GNNs) (Liu et al. 2024), specifically graph attention networks (GAT) (Shaolong and Guo 2024), which employ adaptive neighbor-weighting to effectively model node interactions in heterogeneous topologies. Despite these advances, the integration of GAT and DRL is still in its infancy, especially for complex multi-hop RPL routing tasks, lacking a mature framework and system verification.

Hence, this paper proposes a novel graph attention deep reinforcement learning-based RPL routing optimization algorithm (GATR). GATR employs an Actor-Critic architecture with centralized training with decentralized execution (CTDE), delivering three key innovations: (1) A dual-channel modeling framework integrating GAT mechanisms with edge matrix encoding in the Critic network, enabling dynamic neighbor-state aggregation and explicit physical-layer interference representation to address partial observability in multi-agent environments; (2) A lightweight three-layer convolutional Actor network incorporating contrastive learning to maintain feature representation while substantially reducing parameter overhead; and (3) A variance-sensitive multi-objective optimization strategy employing dynamic volatility assessment for adaptive weight allocation, ensuring stable QoS under complex network conditions. Based on these, GATR can deliver efficient routing decisions for resource-constrained IoT devices while significantly enhancing protocol adaptability and energy efficiency in dynamic environments. The main contributions of this paper are summarized as follows:

- We first integrate the GAT with DRL for RPL protocol optimization, featuring an innovative topology-aware mechanism. It can enable adaptive neighbor-relationship modeling and provide structure-aware optimization guidance for routing strategies, substantially improving environmental adaptability.
- We propose a novel node-level topology attention mechanism addressing partial observability in multi-agent environments through joint node-state and network-topology representations, significantly enhancing routing decision context-awareness. Meanwhile, we design a lightweight Actor network that employs a three-tier convolutional architecture with parameter sharing to maintain performance while dramatically reducing computational complexity.
- We design an original variance-sensitive multi-objective reward function incorporating both key performance indicators (data transmission reliability, energy balance, and delay) and coefficient of variation-based volatility quantification. Combined with action entropy regularization and contrastive learning mechanisms, it can enhance

training stability and policy diversity while maintaining superior convergence in complex networks.

- We verify the convergence and effectiveness of the GATR algorithm through systematic simulation experiments. Experimental results demonstrate that compared with the current mainstream RPL optimization schemes, the GATR algorithm shows significant advantages in energy efficiency, network lifetime, queue stability, data transmission reliability, and other key performance indicators.

The remainder of this paper is organized as follows: Section 2 reviews related works; Section 3 introduces the system model and problem formulation. Section 4 details GATR architecture and key algorithmic components; Section 5 presents the experimental setup and analyzes results and performance; and Section 6 concludes with future research directions.

2 Related works

This section investigates the state-of-the-art routing optimization algorithms, including traditional RPL routing optimization methods and intelligent learning-based RPL routing optimization algorithms

2.1 Traditional RPL optimization algorithms

Traditional RPL optimization has primarily focused on enhancing the objective function (OF). The minimum rank with hysteresis objective function (MRHOF) algorithm (Gnawali et al. 2012) minimizes routing metrics and introduces a hysteresis mechanism to reduce the frequency of route switching. However, it exhibits poor adaptability under dynamic network topologies. Baghani and Khabbazian (2022) proposed a compatible solution to alleviate the point-to-point path stretch issue in RPL, significantly reducing end-to-end delay and power consumption, though its stability remains limited in large-scale mobility or sudden traffic surge scenarios. Fazli and Mansubbassiri (2022) adopted the analytic hierarchy process (AHP) and VlseKriterijumska optimizacija I kompromisno resenje (VIKOR) multi-criteria decision-making techniques to select parent nodes based on multiple metrics such as link quality and energy. Although it improves routing accuracy, the high computational complexity limits its scalability to large networks. Park et al. (2023) proposed power line communication objective function (PLC-OF) for power line communication (PLC) environments, leveraging physical layer diversity to enhance robustness and channel efficiency, making it suitable for congested scenarios. However, its performance degrades under severe electrical interference. To improve path reliability and energy efficiency, Abbou and Manner

(2023) introduced the expected transmission count remaining energy (ETXRE) metric based on MRHOF, jointly considering ETX and residual energy, significantly enhancing path selection quality, although inaccuracies in energy estimation can diminish its effectiveness. Shahbakhsh et al. (2023) proposed an enhanced RPL protocol, which selects parent nodes based on a reliability assessment mechanism, effectively improving network throughput and topology stability, albeit with increased computational overhead in large-scale networks. To address node mobility scenarios, Hussain and Roopa (2023) designed the best-effort RPL protocol, incorporating a received signal strength indicator-based-based mobility awareness mechanism and load-balancing strategy, thereby enhancing energy efficiency and link stability, making it suitable for dynamic environments. Mishra and Khatua (2024) employed a multipath mechanism to optimize buffer and channel packet loss rates, while introducing backup paths to enhance reliability, making it suitable for mission-critical IoT (MC-IoT) applications, though at the cost of potential resource redundancy. Although traditional optimization methods have improved RPL performance to some extent, most approaches rely on fixed rules and static parameter settings, resulting in limited adaptability to highly dynamic and frequently changing IoT network environments.

2.2 Intelligent Learning-based RPL optimization algorithms

To overcome the limitations of traditional methods, researchers have recently introduced machine learning (ML) and deep learning (DL) techniques to enhance the intelligence of RPL. Specifically, Santos et al. (2023) employed ML to enable dynamic metric adjustment, thereby improving network throughput and energy efficiency. Similarly, Mezher et al. (2024) evaluated various ML algorithms in smart grid networks, revealing that node distribution significantly affects model performance. In addition, Wakili and Bakkali (2024) proposed an adaptive objective function (AOF) that leverages real-time monitoring and OF switching mechanisms to enhance overall performance. Furthermore, Maheshwari and Panneerselvam (2024) combined fuzzy AHP and technique for order of preference by similarity to ideal solution (TOPSIS) to design a multi-criteria routing algorithm, ensuring both load balancing and forwarding efficiency. Mokrani et al. (2025) developed a lightweight energy-aware objective function. It integrates with long short-term memory-based prediction and particle swarm optimization optimization, achieving significant improvements in energy efficiency and stability. In addition, Cheppali and Selvakumar (2025) introduced a multi-metric fusion optimization model to select optimal parent nodes, thereby effectively enhancing RPL's energy performance. Vidhya et al. (2025) designed an environment-aware power predic-

tion mechanism to dynamically adjust transmission power, thus improving RPL robustness against external interference.

Recently, RL-based approaches for RPL optimization have made notable progress. Farag and Stefanovič (2021) leveraged Q-learning to enhance the nodes' adaptive load-balancing capability. Lalani et al. (2024) integrated node mobility and link quality metrics into the Q-learning to improve routing stability. Zahedy et al. (2024) proposed the three-phase reinforcement learning-based RPL protocol, which refines parent and path selection strategies. Dey and Ghosh (2024) introduced the intelligent and trusted RPL framework, incorporating a behavioral trust mechanism and multi-agent reinforcement learning (MARL) to strengthen RPL's resilience against internal distributed attacks. Wang et al. (2023) constructed a dual-attention deep reinforcement learning model named DADR, which combines multi-scale convolutional attention with multi-head self-attention modules to significantly optimize energy efficiency and data reliability. Lei and Liu (2024) proposed the RARL algorithm based on dueling double deep Q network, adopting a centralized training and distributed execution architecture to alleviate energy consumption and load imbalance issues in large-scale IoT networks.

Despite recent advances in intelligent RPL optimization, several challenges remain, i.e., most approaches focus on a single performance metric, lacking dynamic trade-offs among energy efficiency, reliability, and latency. Moreover, their adaptability to highly dynamic or uncertain networks is limited, resulting in poor responsiveness to topology changes. In addition, the high training and computational costs of some methods hinder deployment on resource-constrained devices. To address these issues, we propose a GAT-enhanced DRL algorithm for RPL. By integrating GAT mechanisms to better capture topological features and leveraging RL for multi-metric optimization, the proposed method significantly improves adaptability and robustness in dynamic IoT environments.

3 System model and problem formulation

3.1 Network model

We consider a low-power and vulnerable network (LLN) composed of N static nodes. All nodes are distributed in a two-dimensional geographical region, which can be modeled as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$ represents the node set. The $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ indicates the set of communication links of nodes in the networks. The sink node v_0 is responsible for collecting data from other nodes, and the other nodes periodically report monitoring data to it in the many-to-one communication mode. All nodes have

sensing, computing, and communication capabilities, but are limited to resource constraints (i.e., limited battery energy E_i^{init} , fixed communication radius r (only nodes within the range are allowed to communicate directly), and limited processing and storage capabilities. The communication link of nodes is asymmetric and time-varying, and its quality is dynamically affected by environmental interference, distance, multipath fading, and other factors, resulting in the link reliability and energy consumption level changing with time. Let $(v_i, v_j) \in \mathcal{E}$ denote the directed link between nodes v_i and v_j , if and only if v_j is within the communication radius of v_i and the link quality meets certain threshold requirements. The RPL protocol constructs a destination-oriented directed acyclic graph (DODAG) by having each node select an optimal parent through an objective function.

The energy consumption of nodes mainly comes from the data transmission process. This paper utilizes a first-order energy consumption model, which has been widely used in several IoT environments (Kaur et al. 2021). Based on this, the energy consumption E_{tx} for transmitting b -bit data with distance d is given by:

$$E_{tx} = \begin{cases} bE_{elec} + bE_{fs}d^2, & d \leq d_0 \\ bE_{elec} + bE_{mp}d^4, & d > d_0 \end{cases}, \quad (1)$$

where E_{elec} is the circuit operation consumes basic energy; d_0 is the transmission distance threshold; E_{fs} and E_{mp} are the required amplification energy. $\tau \in [2, 4]$ is the path loss factor. The receiving energy consumption E_{rx} is defined as:

$$E_{rx} = pE_{elec}. \quad (2)$$

The residual energy of node v_i in the t -th period $E_{i,r}^t$ is:

$$E_{i,r}^t = E_{i,r}^{t-1} - \sum_{j \in N_i^f} E_{tx} - \sum_{k \in N_i^r} E_{rx}, \quad (3)$$

where N_i^f is the number of data forwarded by node v_i , and N_i^r is the number of data received from child nodes. The node $v_i \in \mathcal{V}$ periodically generates data packets at the rate λ_i .

Let D_{prop} , D_{tran} , D_{proc} , and D_{queue} denote the propagation delay, data transmission delay, processing delay, and queuing delay, respectively. Accordingly, the network communication delay of node v_i is defined as:

$$DE_i^t = D_{i,prop}^t + D_{i,tran}^t + D_{i,proc}^t + D_{i,queue}^t. \quad (4)$$

To simplify the analysis, we assume that the traffic generation process follows the Poisson distribution, and all nodes have a maximum cache queue capacity of q_{max} . Congestion occurs when the queue is full or the link packet loss rate increases. The network adopts the time-sharing transmission mechanism, and the nodes use the first-come first-served strategy to carry out multi-hop forwarding according to the data path established by the improved RPL protocol.

3.2 Problem formulation

The optimization problem of selecting forwarding nodes F^t for network nodes, aiming to maximize the network lifetime and transmission reliability, is formulated as:

$$\max_{F^t} \sum_{i=1}^N \omega_e^t E_{r,i}^t + \omega_d^t D E_i^t + \omega_p^t P D R_i^t, \quad (5)$$

Subject to

$$\exists (v_i, v_j) \in \mathcal{E}, \forall v_i, v_j \in \mathcal{V} \setminus \{v_0\}; \quad (5a)$$

$$\sum_{j:(v_i, v_j) \in \mathcal{E}} f_{ij}^t = 1, \forall v_i \in \mathcal{V} \setminus \{v_0\}; \quad (5b)$$

$$E_{i,r}^t \geq E_{\min}, \forall v_i \in \mathcal{V} \setminus \{v_0\}; \quad (5c)$$

$$P D R_i^t \geq P D R_{\gamma}, \forall v_i \in \mathcal{V} \setminus \{v_0\}; \quad (5d)$$

$$D E_i^t \leq \Delta_{\text{delay}}, \forall v_i \in \mathcal{V} \setminus \{v_0\}; \quad (5e)$$

$$h_i^t < h_j^t, f_{ij}^t = 1, \forall v_i, v_j \in \mathcal{V} \setminus \{v_0\}; \quad (5f)$$

Here, $F^t = [f_{i,j}^t]_{i=1}^N$ is the set of the forwarding node for all nodes in the t -th period. Specifically, $f_{i,j}^t \in \{0, 1\}$ is a binary variable. If node v_j is the forwarding node of node v_i in the t -th period, we have $f_{i,j}^t = 1$ and $f_i^t = j$; otherwise, $f_{i,j}^t = 0$. The w_e^t , w_d^t , and w_p^t are the weight coefficients, respectively. In addition, the constraints (5a) and (5b) are network connectivity constraints and single parent constraints, respectively. Constraints (5c)–(5e) are the minimum residual energy, minimum PDR and maximum delay constraints, respectively, where $E_{\min}$, $P D R_{\gamma}$, and Δ_{delay} denote the minimum residual energy threshold, the minimum PDR, and the maximum allowable transmission delay. To ensure network robustness and avoid premature node failure, the residual energy of each participating node must not fall below the threshold $E_{\min}$. Furthermore, only communication links satisfying the minimum PDR constraint (5d) and the maximum delay constraint (5e) are considered valid. The constraints (5f) is an acyclic routing constraint, used to ensure that the topology constructed is a DODAG, where h_i^t and h_j^t are the hop counts of node v_i and node v_j , respectively. These constraints jointly ensure that selected routes are both energy-efficient and meet quality-of-service (QoS) requirements in terms of reliability and latency.

4 GATR algorithm design

This optimization problem in (5) is a typical nonlinear multi-objective combinatorial optimization problem. The objective function contains nonconvex terms (such as PDR in product form), path dependencies (end-to-end indicators), and the constraints contain discrete variables (node selection, path

constraints), which makes it difficult to solve the problem by traditional optimization methods. Therefore, this paper proposes a distributed routing strategy optimization method based on the GATR algorithm.

4.1 Partially observed markov decision process

This paper formulates the RPL routing optimization problem with multi-objective optimization as a Partially Observed Markov Decision Process (POMDP), denoted as a tuple $M = (\Omega, S, A, P, r, \gamma)$. Let H denote the number of candidate forwarding nodes, and W denote the dimension of considered routing metrics.

Obsevation space Ω Let $\mathbf{o}_i^t \in \mathbb{R}^{3 \times H \times W} \in \Omega$ denote the local observation state of node v_i in the t period. $\mathbf{o}_i^t$ includes the historical traffic feature and link status characteristic of the neighbor nodes of node i , as:

$$\mathbf{o}_i^t = \begin{bmatrix} m_1^t & \cdots & m_j^t & \cdots & m_H^t \\ m_1^{t-1} & \cdots & m_j^{t-1} & \cdots & m_H^{t-1} \\ m_1^{t-2} & \cdots & m_j^{t-2} & \cdots & m_H^{t-2} \end{bmatrix}, \quad (6)$$

where $m_j^t = [h_j^t, q_j^t, E_{r,j}^t, E T X_j^t, N_{l,j}^t]$ is the routing metrics (i.e., hop count, buffer queue length, residual energy, ETX, and the number of potential child nodes) of node v_j and $(v_i, v_j) \in \mathcal{E}$.

State space S $s_t \in S$ is the global state of the t -th period, as:

$$s_t = \{\mathbf{o}_1^t, \dots, \mathbf{o}_N^t\}, \quad (7)$$

where s_t is composed of the local observations of all nodes.

Action space A $\mathbf{a}_i^t \in A$ is the selected action of agent $i \in \mathbb{N}$ at t -th period, denoted as $\mathbf{a}_i^t = [\mathbf{a}_{i,1}^t, \mathbf{a}_{i,2}^t, \dots, \mathbf{a}_{i,H}^t]$. The $\mathbf{a}_{i,h}^t$ is the binary indicator variable, where $\mathbf{a}_{i,h}^t \in \{0, 1\}$ and $h \in H$. If $\mathbf{a}_{i,h}^t = 1$, the h -th node in $\mathbf{o}_{ho,i}^t$ will be selected as its forwarding node for the upcoming data transmission.

Transition function P P denotes the probability distribution over its next local observation state $\mathbf{o}_i^{t+1}$ for a given $\mathbf{o}_i^t$ and $\mathbf{a}_i^t$, i.e., $P = Pr(\mathbf{o}_i^{t+1} | \mathbf{o}_i^t, \mathbf{a}_i^t)$.

Variance sensitive multi-objective reward function R R is the multi-objective reward function that maps the local observation state $\mathbf{o}_i^t$ and action $\mathbf{a}_i^t$ to a reward value. Specifically, if $f_{i,j}^t = 1$ and $j \in \mathcal{N}_i^t$, the r_i^t for $(\mathbf{o}_i^t, \mathbf{a}_i^t)$ is defined as:

$$r_i^t = w_e^t r_{e,i}^t + w_d^t r_{d,i}^t + w_p^t r_{p,i}^t, \quad (8)$$

where $\mathbf{r}_{e,i}^t = E_{r,i}^t$, $\mathbf{r}_{d,i}^t = DE_{avg}/DE_i^t$, and $\mathbf{r}_{p,i}^t = PDR_i^t$ are the subobjective reward values of residual energy, end-to-end delay, and PDR of node i in GATR algorithm, respectively. The DE_{avg} is the average end-to-end delay.

To achieve dynamic weight allocation for w_e^t , w_d^t , and w_p^t , we propose a variance-sensitive multi-objective reward weight optimization mechanism with a sliding window adaptive coefficient of variation (CoV) smooth weighting method. Based on this, the $\mathbf{r}_i^t$ can suppress the excessive impact of high variance objectives on strategy optimization. Specifically, the monitoring window size is adjusted based on network dynamics intensity ΔCoV_{avg} , as:

$$W' = W_{base} + \beta_W |\Delta CoV_{avg}^t|, \quad (9)$$

where the base window ($W_{base} = 10$) captures bursty fluctuations, β_W balances computational constraints with responsiveness, and $|\Delta CoV_{avg}^t|$ is the absolute difference of the average CoV between adjacent windows, used to quantify the dynamic strength of the network. The CoV of k -subobjective during given windows W' is:

$$CoV_k^t = \sigma_k^2 / \mu_k, \quad (10)$$

where CoV_k^t is used to quantify the relative volatility of this indicator. Within window W' , each objective's mean (μ_k) and variance (σ_k) can be computed. The weighting uses nonlinear decay, as:

$$w_k' = 1 / (1 + CoV_k^t), \quad (11)$$

where $k \in \{e, d, p\}$. Based on this, the w_k^t can be updated by:

$$w_k^t = \alpha_w w_k' + (1 - \alpha_w) w_0, \quad (12)$$

where α_w is the temperature coefficient used to control the mixing ratio of w_k' and $w_0 = 1/3$ weights.

Note that if $f_{i,j}^t = 1$ and $(v_i, v_j) \notin \mathcal{E}$, we have $\mathbf{r}_i^t = -1$. To ensure data transmission reliability, the action of node i is updated as $a_i^t = a_i^{t-1}$.

Discount factor γ $\gamma \in (0, 1]$ is the discount factor for trading off the immediate and future rewards.

4.2 GATR algorithm

4.2.1 GATR architecture

This paper proposes a graph attention-based deep reinforcement learning architecture, i.e., GATR, under the actor-critic

(AC) framework, which achieves optimization decision-making for the RPL routing protocol by integrating multi-scale convolution with a GAT mechanism. As shown in Fig. 1, the model adopts the CTDE paradigm, and its core architecture comprises three key components: 1) a decentralized and lightweight Actor network serving as the local decision-making model for nodes; 2) a global Critic network based on the GAT mechanism that provides scalable value evaluation; 3) and a hybrid training mechanism combining contrastive learning with action entropy regularization.

The AC framework with CTDE is particularly well suited for energy-constrained IoT routing scenarios. Its single-step policy updates reduce computational and memory overhead, while the CTDE paradigm allows for centralized Critic training alongside lightweight decentralized Actors, thereby minimizing communication costs. Compared to proximal policy optimization algorithms, which rely on batch updates and impose greater computational demands, the AC with CTDE paradigm provides improved efficiency and adaptability in dynamic IoT environments. Furthermore, by separating policy execution from value evaluation and leveraging global information through the Critic network, this architecture enhances convergence speed and cross-scenario generalization, all within the resource limitations typical of IoT nodes.

4.2.2 Actor network

To address the resource constraints inherent in IoT nodes, the GATR designs a lightweight convolutional Actor network, as illustrated in Fig. 2. This network comprises only three convolutional layers and is specifically tailored to efficiently extract local observations from graph-structured node data. By balancing model performance and computational cost, it is particularly suitable for low-power and large-scale IoT scenarios.

Specifically, the input to the Actor network at time step t is the local observation of node i , denoted as $o_i^t \in \mathbb{R}^{3 \times H \times W}$, where H is the number of candidate forwarding nodes and W is the dimension of routing metric. The input tensor integrates multiple routing indicators, including link quality, residual energy, and queue status.

First, the input is passed through a $C_a^1 \times 1 \times 1$ convolutional layer for channel compression and feature mapping:

$$X_1 = \text{ReLU} \left(\text{Conv}^{C_a^1 \times 1 \times 1} (o_i^t) \right), \quad (13)$$

where C_a^1 is the number of filters in the first convolutional layer, and $\text{ReLU}(\cdot)$ is the activation function. The 1×1 convolution performs a linear transformation on the multidimensional inputs without altering the spatial structure, effectively reducing redundancy and improving computational efficiency.

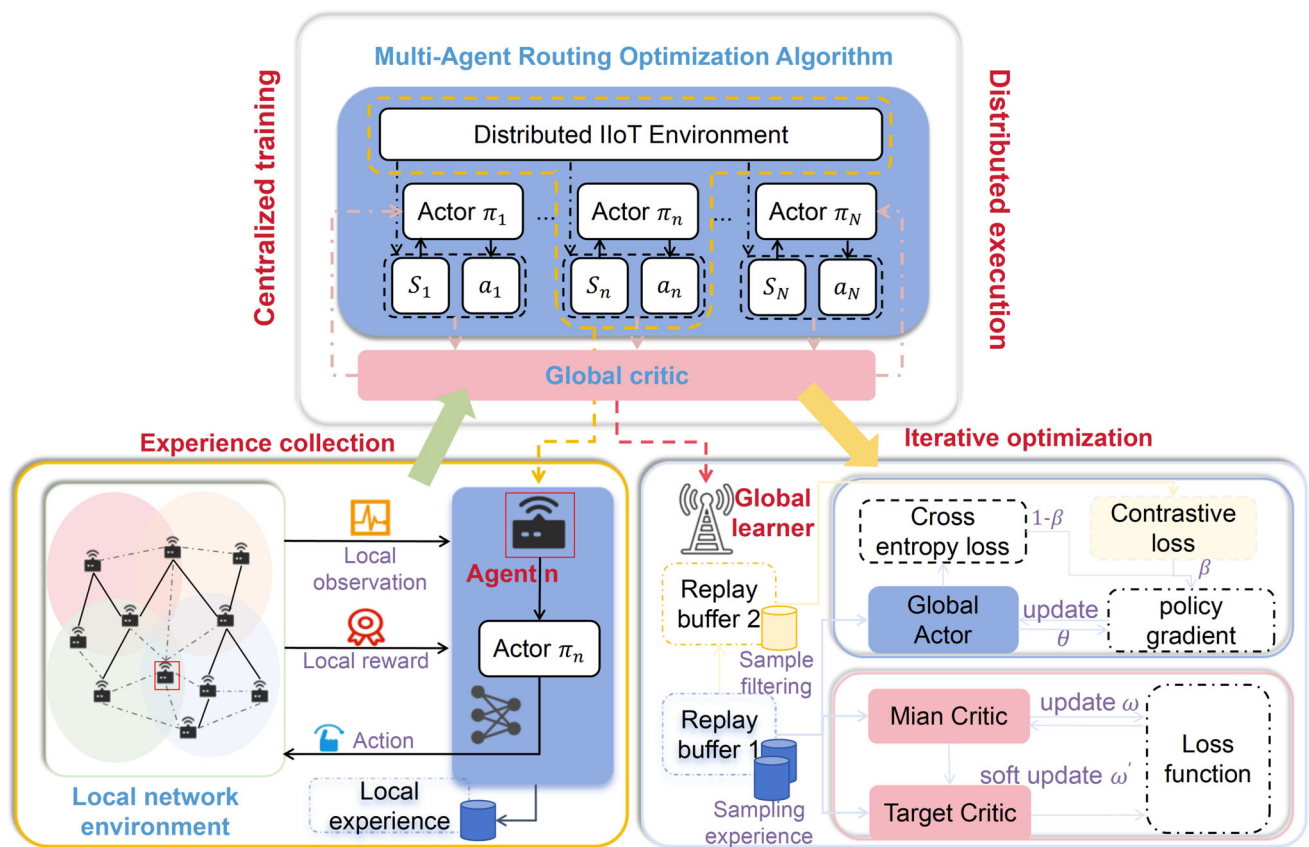


Fig. 1 GATR based on CTDE architecture

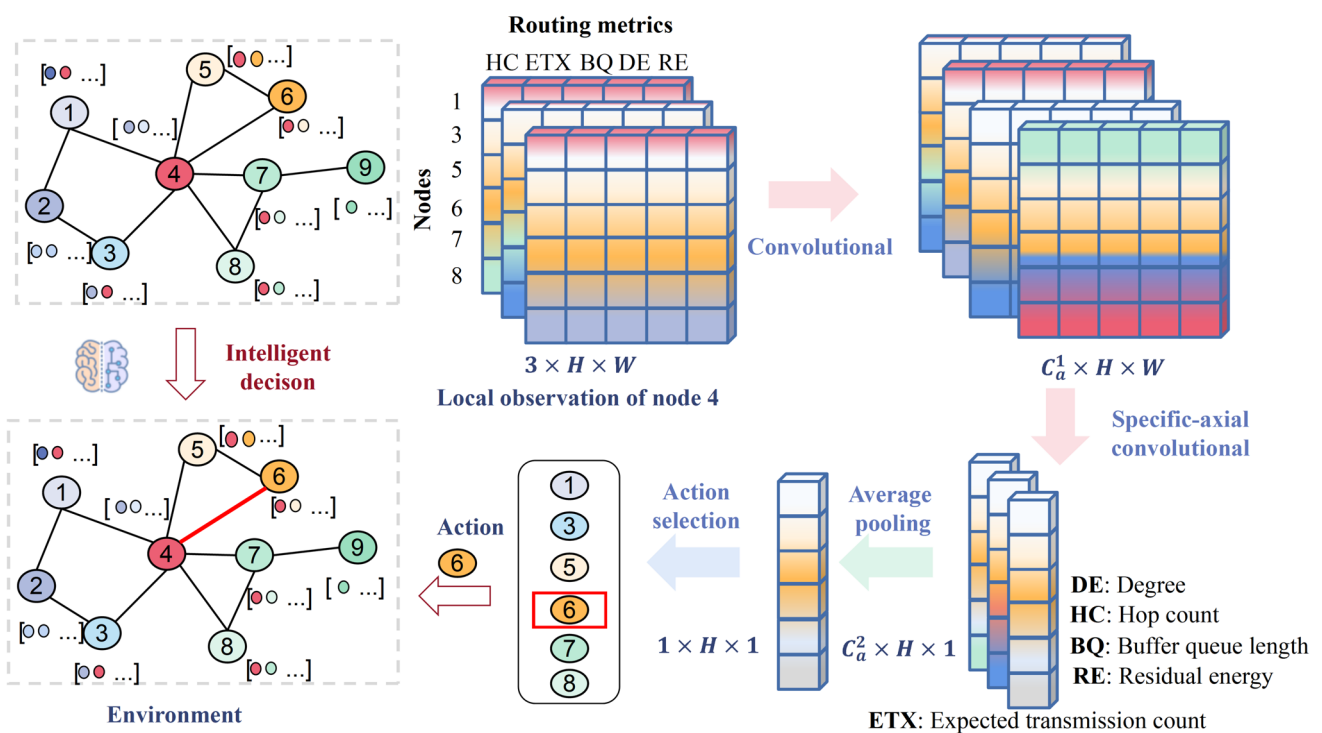


Fig. 2 Actor network for data forwarding decision

Next, a convolutional layer with a sliding window of size $1 \times W$ is used to extract the mixed features of candidate parent nodes and capture inter-node dependencies:

$$X_2 = \text{ReLU} \left(\text{Conv}^{C_2^q \times 1 \times W}(X_1) \right), \quad (14)$$

where C_2^q is the number of filters in the second layer. This layer enhances the model's capacity to capture correlations among the routing metrics of different candidates through local receptive fields. Finally, a $1 \times 1 \times 1$ convolution is applied to generate a probability distribution over possible actions:

$$X_3 = \text{Softmax} \left(\text{Conv}^{1 \times 1 \times 1}(X_2) \right), \quad (15)$$

where the $\text{Softmax}(\cdot)$ function normalizes the output to form a valid probability distribution, guiding the node to select the optimal forwarding path.

The proposed lightweight convolutional Actor network supports end-to-end learning of multi-dimensional routing metrics without manual weighting, thereby avoiding subjective bias and enhancing decision objectivity. Meanwhile, by leveraging local convolutional receptive fields, it autonomously learns the spatial correlations among candidate forwarding nodes, overcoming the limitations of tradi-

tional RPL algorithms that rely solely on single-hop metrics. Moreover, the translation-invariance property of convolutional operations enables the model to maintain robustness and adaptability under dynamic network topologies, such as node addition or removal. Compared to fully connected structures, the convolutional design significantly reduces computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$, making it particularly suitable for real-time routing decisions in large-scale IoT environments.

4.2.3 GAT-based critic network

To address the limitation of partial observability in multi-agent systems while minimizing computational complexity, GATR introduces a global Critic network based on the GAT mechanism, as illustrated in Fig. 3. The multi-agent environment is modeled as a graph, where each node corresponds to an agent, and its local observation is embedded as the node feature. Communication relationships among agents are encoded via an adjacency matrix. The Critic network aggregates local observations through the GAT structure to learn a global state-value function under the joint policy. By dynamically assigning attention weights during information aggregation, the Critic network can focus on neighbors most relevant to policy decisions, thereby guiding individual

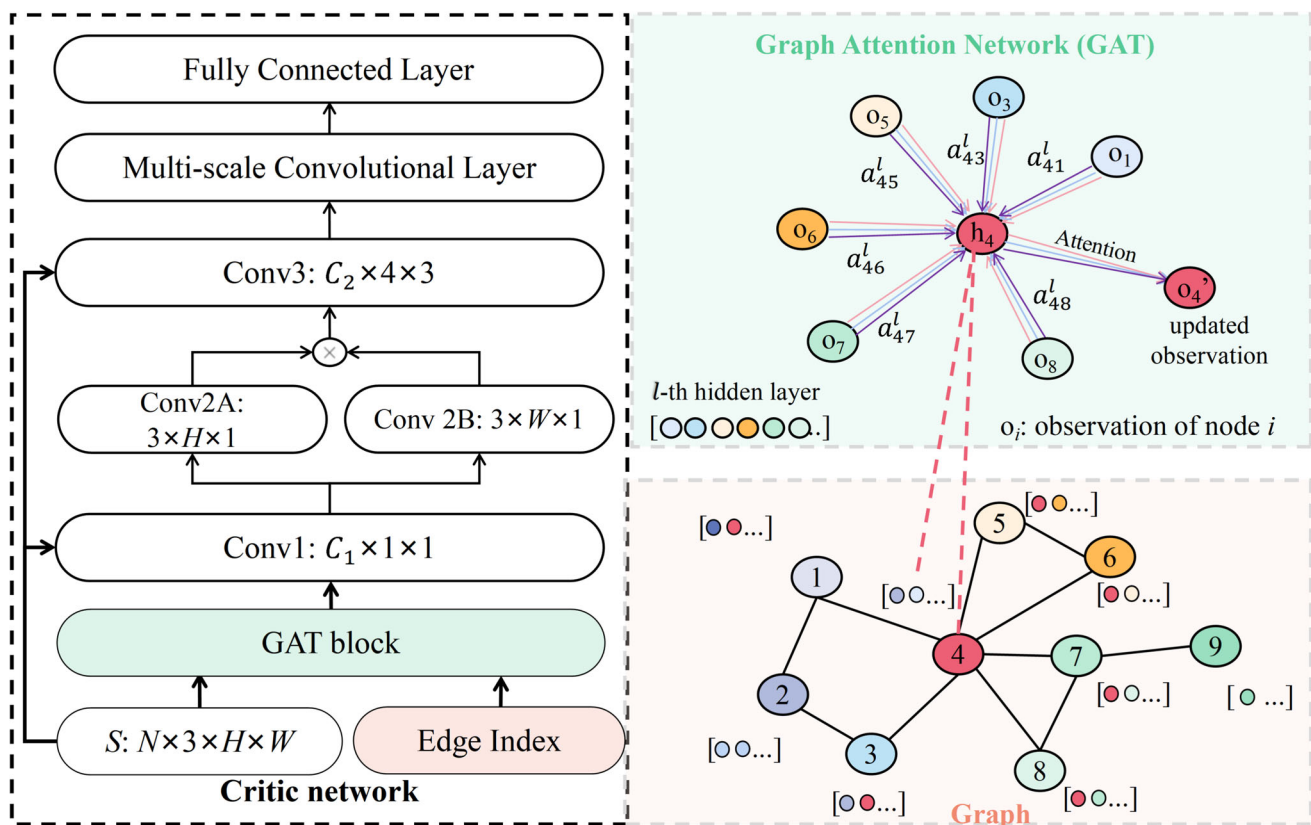


Fig. 3 GAT-based Critic Network

Actor networks toward more cooperative and globally optimal strategies while mitigating the impact of limited local observability.

To transform the input feature o_i^t into higher-level features, the Critic network first applies a shared linear transformation to each node, parameterized by a weight matrix:

$$\mathbf{h}_i = \mathbf{W}o_i^t, \quad (16)$$

where $\mathbf{W} \in \mathbb{R}^{F' \times F}$. For each neighbor $v_j \in \mathcal{N}_i$, a self-attention mechanism $a : \mathbb{R}^{F'} \times \mathbb{R}^{F'} \rightarrow \mathbb{R}$ is employed to compute attention coefficients:

$$e_{ij} = att(\mathbf{h}_i, \mathbf{h}_j) = \text{LeakyReLU}(\tilde{\mathbf{a}}^T [\mathbf{h}_i \| \mathbf{h}_j]), \quad (17)$$

where $\tilde{\mathbf{a}} \in \mathbb{R}^{2F'}$ is a learnable parameter, and $[\cdot \| \cdot]$ denotes vector concatenation. To make coefficients easily comparable across different nodes, we normalize them across all choices of v_j using the softmax function:

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik})}, \quad (18)$$

The attention mechanism att is a single-layer feedforward neural network parametrized by a weight vector $\tilde{\mathbf{a}} \in \mathbb{R}^{2F'}$, and applying the LeakyReLU nonlinearity. Fully expanded out, the coefficients computed by the attention mechanism may then be expressed as:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\tilde{\mathbf{a}}^T [\mathbf{W}\tilde{\mathbf{h}}_i \| \mathbf{W}\tilde{\mathbf{h}}_j]\right)\right)}{\sum_{k \in \mathcal{N}_i} \exp\left(\text{LeakyReLU}\left(\tilde{\mathbf{a}}^T [\mathbf{W}\tilde{\mathbf{h}}_i \| \mathbf{W}\tilde{\mathbf{h}}_k]\right)\right)}, \quad (19)$$

where T represents transposition and $\|$ is the concatenation operation. Based on this, the updated feature of node v_i is computed as a weighted sum of its neighbors' features:

$$\tilde{\mathbf{h}}'_i = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{h}_j\right), \quad (20)$$

where $\sigma(\cdot)$ denotes a non-linear activation function.

To further improve the model's expressiveness and stability, the Critic network employs a multi-head attention mechanism. Specifically, K independent attention heads are computed in parallel. The average pooling is used instead of concatenation to aggregate multiple attention heads, and the final activation is deferred:

$$\tilde{\mathbf{h}}'_i = \sigma\left(\frac{1}{K} \sum_{k=1}^K \sum_{j \in \mathcal{N}_i} \alpha_{ij}^{(k)} \mathbf{W}^{(k)} \tilde{o}_j\right), \quad (21)$$

where $\alpha_{ij}^{(k)}$ are normalized attention coefficients computed by the k -th attention mechanism ($a^{(k)}$), and $\mathbf{W}^{(k)}$ is the corresponding input linear transformation's weight matrix.

As shown in Fig. 3, taking node h_4 as an example, it aggregates local observations from neighboring nodes $o_1, o_3, \dots, o_8$, each associated with attention weights $a_{41}^1, a_{43}^1, \dots, a_{48}^1$, to produce the updated feature representation o'_4 . This attention mechanism effectively models dependencies among nodes and promotes efficient propagation of local information toward a global representation.

Subsequently, the enhanced feature vector $\tilde{\mathbf{h}}'_i$ is reshapes as $\tilde{\mathbf{h}}'_i \in \mathbb{R}^{3 \times H \times W}$. Then, $\tilde{\mathbf{h}}_i^2 = \tilde{\mathbf{h}}'_i + o_i^t$ is used as the input to the next parallel convolutional layer with two different filter kernels $\{C \times 1 \times H\}$ and $\{3 \times W \times 1\}$ to obtain $\tilde{\mathbf{h}}_i^3$, as:

$$\tilde{\mathbf{h}}_i^3 = \text{Conv}^{3 \times 1 \times W}(\tilde{\mathbf{h}}_i^2) \cdot \text{Conv}^{3 \times H \times 1}(\tilde{\mathbf{h}}_i^2), \quad (22)$$

where the parallel convolutional layer aims to extract the hidden important information from the node domain and resource domain of $\tilde{\mathbf{h}}_i^2$, thereby improving the feature representation ability. Moreover, GATR incorporates a multi-scale convolution (MSC) structure within the Critic network to further enhance feature expressiveness. Specifically, four different filter kernels in set $\mathbb{K} = \{1 \times 1 \times 1, 1 \times 1 \times 2, 1 \times 2 \times 1, 1 \times 2 \times 2\}$ are designed to extract the hidden important information from different receptive fields in $\tilde{\mathbf{h}}_i^3 \in \mathbb{R}^{3 \times 3 \times 3}$. It utilizes multiple convolutional kernels with different receptive fields to extract features at various spatial scales. Such integration of convolutional and attention mechanisms enables the network to capture both fine-grained local and global structural information.

4.2.4 GATR-based data forwarding

Algorithm 1 outlines the distributed interaction process for node $v_i \in \mathcal{V}$ in the GATR framework. During the control phase (CP), node i acquires its local observation o_i^t and selects an action a_i^t by $\mathbf{a}_i^t = \arg \max_{\mathbf{a} \in \mathcal{A}} \{\pi_{\phi_i}(\mathbf{a} | \mathbf{o}_i^t)\}$, where $\pi_{\phi_i}(\mathbf{a} | \mathbf{o}_i^t)$ denotes the local Actor network with parameters ϕ_i . These parameters are periodically synchronized with the centralized learner's global parameters ϕ . Upon executing a_i^t , the environment returns an instantaneous reward r_i^t (defined in (8)) and transitions to the next state o_i^{t+1} (6). Node v_i then constructs an experience tuple $\mathbf{e} = \langle \mathbf{o}_i^t, \mathbf{a}_i^t, r_i^t, \mathbf{o}_i^{t+1} \rangle$ and transmits it to the centralized learner for storage in the experience replay buffer 1, i.e., $\mathbb{D}$. The selected action a_i^t is mapped to a forwarding node j via $\text{map}(\mathbf{a}_i^t, \mathbf{o}_i^t)$. We enable the validity verification mechanism (Wang et al. 2023) to ensure the reliable data transmission by cross-checking node v_j 's eligibility (e.g., energy reserves and link stability). During the subsequent data transmission phase (DTP), node v_i forwards packets to f_i^t using the carrier

sense multiple access with collision avoidance (CSMA/CA) protocol, ensuring collision-free communication in shared channels.

Algorithm 1 : GATR-based Data Forwarding Mechanism.

Require: Local observation o_i^t
Ensure: Action a_i^t and forwarding node f_i^t

```

1: for each CP period do
2:   Update actor parameters:  $\phi_i \leftarrow \phi$ 
3:   Receive reward  $r_i^{t-1}$  for  $(o_i^{t-1}, a_i^{t-1})$ 
4:   Observe new local state  $o_i^t$ 
5:   Store experience  $e \leftarrow \langle o_i^{t-1}, a_i^{t-1}, r_i^{t-1}, o_i^t \rangle$ 
6:   Select action  $a_i^t \leftarrow \arg \max_{a \in \mathcal{A}} \pi_\phi(a|o_i^t)$ 
7:   Map to forwarding node  $j \leftarrow \text{map}(a_i^t, o_i^t)$ 
8:   if  $(v_i, v_j) \in \mathcal{E}$  then
9:      $f_i^t \leftarrow j$ 
10:  else
11:     $f_i^t \leftarrow f_i^{t-1}$ 
12:  end if
13: end for
14: for each data transmission period do
15:   Transmit data to  $f_i^t$  via CSMA/CA mechanism
16: end for
```

4.3 GATR training mechanism

During the training phase, the global learner leverages diverse experiential data collected from distributed nodes to extract broadly applicable knowledge and principles. This rich experience information enables global learners to quickly obtain efficient routing decision strategies, thus speeding up the convergence speed of the model. To further enhance training efficiency and decision-making performance, GATR incorporates contrastive learning to strengthen the representational capacity of the actor network, along with an entropy-based optimization strategy to reduce uncertainty in decision-making.

To improve the decision-making ability of agents, the policy network (Actor) in GATR uses a joint loss function composed of cross-entropy loss and contrast loss to optimize the Actor network training mechanism. At the early stage of training, contrastive loss dominates the learning process, helping the model to capture fine-grained data features. As training progresses, cross-entropy loss becomes the primary driver, optimizing the model's decision-making ability. The objective function of the contrastively enhanced actor network is defined as:

$$\mathcal{L}_\pi(\phi) = L_{CL}^{(c)} + \beta \cdot L_{RL}^{(b)}, \quad (23)$$

where β is a temperature coefficient. $L_{CL}^{(c)}$ represents the contrastive loss used for feature learning, while $L_{RL}^{(b)}$ denotes the cross-entropy loss for classification. Each loss function uses a distinct sampling strategy.

To ensure the accuracy of experiences used for contrastive learning, an experience filtering method based on the advantage function is employed to generate contrastive training samples. Let d_{td} be a binary indicator, defined as:

$$d_{td} = \begin{cases} 1, & \text{if } A(o_i^t, a_i^t) > 0 \\ 0, & \text{otherwise} \end{cases}, \quad (24)$$

where $A(o_i^t, a_i^t) = Q(o_i^t, a_i^t) - V(o_i^t)$ is the advantage function; $Q(o_i^t, a_i^t)$ is the state-action value and $V(o_i^t)$ is the state value. If the advantage is positive, i.e., the action yields a reward higher than the average, the corresponding experience $e_t^i = \langle o_i^t, a_i^t, r_i^t, o_{i+1}^i \rangle$ will be added to buffer $\mathbb{E}_C$ as a positive sample stored in replay buffer 1 for contrastive loss computation.

The parameters of the Actor network ϕ can be updated as:

$$\phi = \phi - \lambda_\pi \nabla_\phi \mathcal{L}_\pi(\phi), \quad (25)$$

where $\lambda_\pi \in (0, 1]$ is the learning rate of actor network.

To alleviate the over-estimation problem in the estimation of value function, GATR employs a double Critic architecture. The loss function of the main critic network is defined as:

$$L_Q(\theta) = \frac{1}{2B} \sum_{i=1}^B \left(Q_\theta(o_i^t, a_i^t) - y_t^i \right)^2, \quad (26)$$

where y_t^i is the target Q-value, defined as:

$$y_t^i = r_t^i + \gamma Q_{\theta'}(o_{t+1}^i, a_{t+1}^i), \quad (27)$$

where θ' denotes the parameters of the target critic network. The action a_{t+1}^i is selected as:

$$a_{t+1}^i = \arg \max_{a \in \mathcal{A}} Q_\theta(o_{t+1}^i, a), \quad (28)$$

where $Q_{\theta'}(\cdot, \cdot)$ is the target Critic network introduced to mitigate overestimation caused by bootstrapping.

To improve policy exploration and learning efficiency, an entropy-augmented reward function is introduced to encourage exploration of diverse policies:

$$\hat{r}_t^i = r_t^i + \gamma \epsilon \mathbb{E} \left[H_\phi \left(\pi \left(\cdot \mid o_{t+1}^i \right) \right) \right], \quad (29)$$

where $H_\phi(\pi(\cdot \mid o_{t+1}^i)) = -\log \pi_\phi(\cdot \mid o_{t+1}^i)$ denotes the entropy term that promotes learning stability and exploration. ϵ is the temperature parameter controlling the entropy weight.

The critic network is updated via stochastic gradient descent:

$$\theta \leftarrow \theta - \lambda_\theta \nabla_\theta L_Q(\theta), \quad (30)$$

where λ_θ is the learning rate and $\nabla_\theta L_Q(\theta)$ is the gradient of the critic loss function.

To stabilize training and avoid model stagnation or convergence to poor local minima due to parameter misalignment, a soft update strategy is employed for updating the target critic network:

$$\theta' \leftarrow \tau\theta + (1 - \tau)\theta', \quad (31)$$

where $\tau \in (0, 1)$ is the update coefficient controlling the rate of blending.

Algorithm 2 : GATR Training Mechanism.

Require: Experience buffer $\mathbb{E}$, $\mathbb{E}_c$, temperature ψ , entropy coefficient ϵ , learning rates λ_θ , λ_ϕ

Ensure: Updated parameters of Actor ϕ and Critic θ

```

1: Initialize Actor network  $\pi_\phi$ , Critic networks  $Q_\theta, Q_{\theta'}$ 
2: for each training iteration do
3:   Sample  $\{(s_t^i, a_t^i, r_t^i, s_{t+1}^i)\}_{i=1}^B$  from  $\mathbb{E}$ 
4:   Set  $d_i \leftarrow 1$  if  $[A(s_t^i, a_t^i) > 0]$ 
5:   Select positive samples  $\mathbb{E}_c$  where  $d_i = 1$ 
6:   Compute  $L_{CL}^{(c)}$  on  $\mathbb{E}_c$ 
7:   Compute  $L_{RL}^{(b)}$  via cross-entropy
8:   Compute  $\mathcal{L}_\pi(\phi) = L_{CL}^{(c)} + \beta \cdot L_{RL}^{(b)}$ 
9:   Update actor:  $\phi \leftarrow \phi + \lambda_\phi \nabla_\phi L_\pi$ 
10:  Compute  $L_Q = \frac{1}{2B} \sum_{i=1}^B (Q_\theta(s_t^i, a_t^i) - y_t^i)^2$ 
11:  Update main Critic:  $\theta \leftarrow \theta - \lambda_\theta \nabla_\theta L_Q$ 
12:  Update target Critic:  $\theta' \leftarrow \varsigma\theta + (1 - \varsigma)\theta'$ 
13: end for

```

4.4 Complexity analysis

The GATR algorithm achieves efficient complexity distribution through a well-designed CTDE architecture. The Actor network adopts a lightweight structure with three convolutional layers ($C_1^a \times 1 \times 1 \rightarrow C_2^a \times 1 \times W \rightarrow 1 \times 1 \times 1$), resulting in a total complexity of $\mathcal{O}(H \cdot W \cdot M)$, where H is the number of candidate nodes, W the feature dimension, and M the sliding window size. This design keeps the number of parameters under 5K, reducing the model size by approximately 72% compared to traditional fully connected networks. The Critic network integrates a GAT and multi-scale convolutional layers to model global coordination among agents. The GAT layer, leveraging the sparse nature of graphs, reduces the complexity from the traditional multi-head self-attention (MHSA) mechanism's $\mathcal{O}(N^2 \cdot d)$ to $\mathcal{O}(|E| \cdot d' + N \cdot d'^2)$, where $|E|$ is the number of edges, d' is the intermediate feature dimension, and d'^2 accounts for multi-head representations in high-dimensional space. The multi-scale convolution module contributes only a linear complexity of $\mathcal{O}(H \cdot \sum_k C^{(k)})$ through parallel branches with different receptive fields, where $C^{(k)}$ represents the output channels of the k -th branch. In addition, the fully connected

layer has a complexity of: $\mathcal{O}(d_f \cdot H)$, where d_f is the number of neurons in fully connected layer. Under the CTDE paradigm, the complex Critic network (approximately 4.7M FLOPs) is executed on a central server during training, while each agent only runs a lightweight Actor network (approximately 1.2M FLOPs) during execution, requiring less than 1MB of memory. By combining a lightweight convolution-based Actor, an efficient Critic exploiting graph sparsity, and a training-inference separation via CTDE, GATR significantly improves computational efficiency, reduces memory usage, and lowers communication overhead.

5 Performance evaluation

5.1 Experimental setup

We conduct the simulation experiments on a custom IoT network simulation platform built with Python 3.7 and PyTorch 1.10, running on hardware with macOS Monterey 12.6, a 2GHz Intel Core i5 processor, 16GB RAM, and Intel Iris Plus

Table 1 Simulation parameters

Parameters	Values
Network area	300m $\times$ 300m
Topology	Random deployment
Number of nodes	[60, 80, 100]
Traffic rates	[60, 90, 120]ppm/node
Data rate	250kbps
Packet length	128Byte
PHY and MAC protocol	IEEE 802.15.4
Transmission ranges	100m
Interference ranges	120m
Mac max retransmissions	4
Time slot duration	5 ms
Buffer size	8
Initial energy	10J
E_{ele}	50nJ/bit
ϵ_{fs}	10pJbit/m ²
ϵ_{mp}	0.0013pJ/bit/m ²
C, H, W	3, 6, 5
C_a^1, C_a^2	16, 3
GAT Heads	4
C_1, C_2	3, 3
Convolutional kernels	(1,1), (2,1), (2,1), (2,2)
Fully connected layer	(25, 6)
$\lambda_\pi, \lambda_\theta, \gamma$	0.003, 0.001, 0.9
Replay buffer capacity	1000
λ_{tc}, τ	50, 0.01

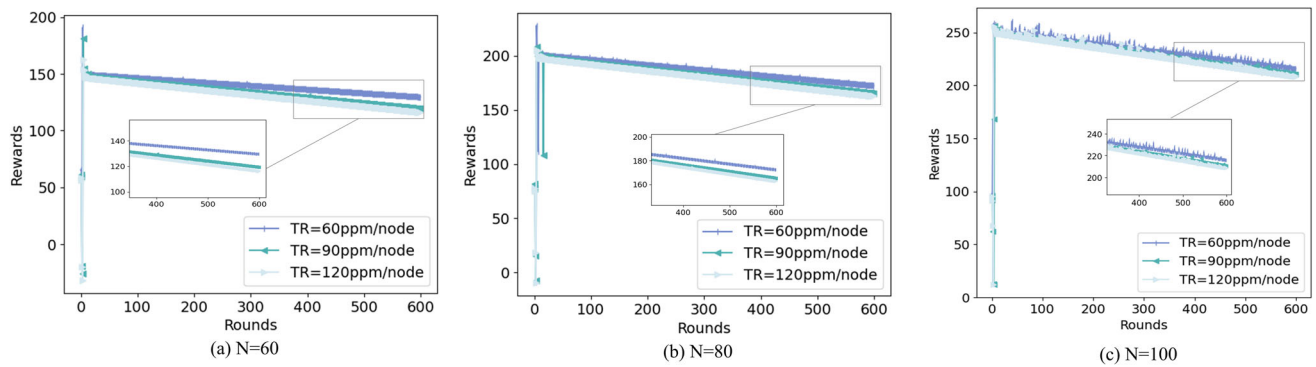


Fig. 4 The convergence results of GATR in different network scenarios

Graphics. All nodes are randomly deployed in a $300\text{m} \times 300\text{m}$ area with the sink node at the center, using the parameters shown in Table 1: node counts (60/80/100), communication radius (100m), traffic load (60/90/120 ppm/node), and channel bandwidth (250kbps). The algorithm hyperparameters include an actor learning rate of 0.003, critic learning rate of 0.001, soft update coefficient τ of 0.01, discount factor γ of 0.9, and 4 graph attention heads. Moreover, we implement four baseline comparison algorithms: (1) traditional RPL with MRHOF (Gnawali et al. 2012), (2) Q-Learning enhanced CARL (Frag and Stefanovič 2021), (3) value-based DDQN and D3QN (RARL) (Lei and Liu 2024), and (4) Actor-Critic with MHSA (DADR) (Wang et al. 2023). All algorithms share identical state space definitions and reward functions, combining latency, energy, and PDR metrics. Meanwhile, we design two validation scenarios: a scalability test with fixed traffic and increasing nodes ($60 \rightarrow 100$), and a load adaptability test with fixed nodes and varying traffic ($60 \rightarrow 120$ ppm/node). The evaluation framework assesses convergence through training stability, individual metrics (end-to-end delay, PDR, network lifetime, and control overhead), and comprehensive performance via multidimensional radar charts.

5.2 Convergence analysis

We conduct multiple comparative experiments to verify the convergence performance of the GATR algorithm under different network scenarios. As shown in Fig. 4, the reward curve of GATR exhibits fluctuations in the initial training phase and then gradually stabilizes (with a final fluctuation amplitude of less than 4.69%). Specifically: 1) Under the high-load condition of 120 ppm/node, the system's final reward value decreases by 3.4% compared to the 60 ppm/node benchmark, in Fig. 4(a). This is because that high-load scenarios intensify the complexity of routing decisions; 2) When expanding from 60 nodes to 100 nodes, the algorithm maintains a highly consistent convergence curve; 3) The decline in the reward value after stabilization (an average decrease of 3.1% every 100 rounds, when $N = 100$, $TR = 120\text{ppm/node}$) is due to the consideration of the node's residual energy indicator in the reward function. As the network continues to operate, the node's energy is continuously consumed, and its residual energy gradually decreases. In addition, the enlarged view shows that the average fluctuation range of reward values in the later training stage (after 300 rounds) is within 2.7%. These results indi-

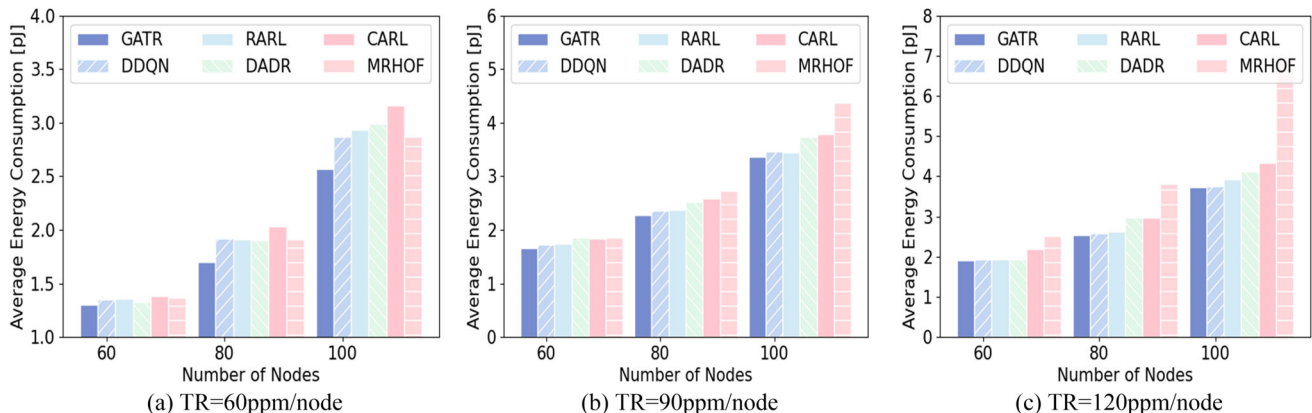


Fig. 5 Comparison of energy consumption on different network scenarios

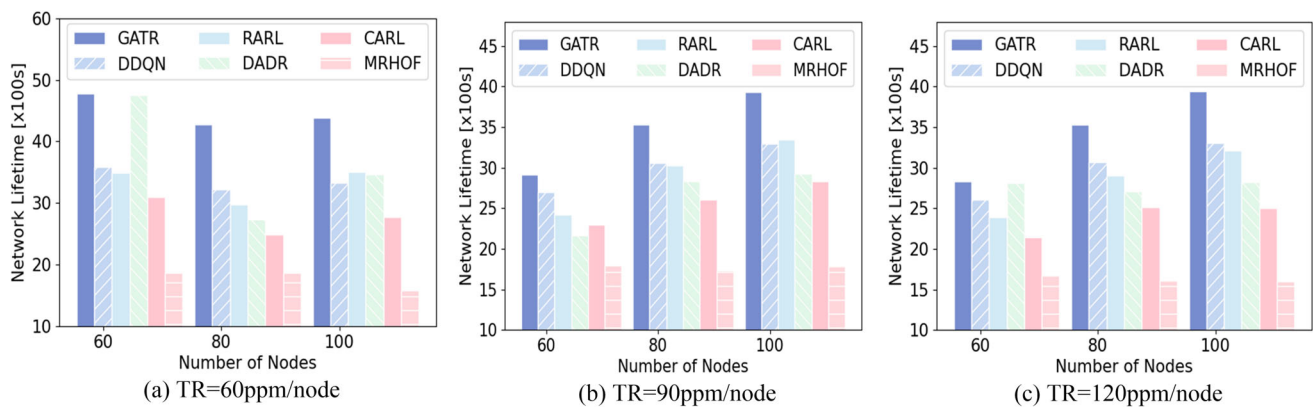


Fig. 6 Comparison of network lifetime in different network scenarios

cate that GATR exhibits stable convergence performance and efficient strategy optimization capabilities when dealing with industrial IoT environments of varying scales and fluctuating traffic loads.

5.3 Network performance

5.3.1 Energy efficiency

Figures 5 and 6 show the average energy consumption performance and network lifetime performance of these algorithms under different traffic rates. As shown in Figs. 5, the GATR algorithm obtains lower average energy consumption in all scenarios. Especially when the number of nodes increases to 100, its energy consumption advantage becomes more significant. Specifically, when $N = 100$, $TR = 60$ ppm/node, the average energy consumption of the GATR algorithm is reduced by about 10.7%, 12.5%, 14.1%, 18.7%, and 10.7% compared to DDQN, RARL, DADR, CARL, and MRHOF, respectively. This indicates that the GATR algorithm can effectively manage energy consumption and maintain high energy efficiency even as network density increases. In contrast, other algorithms, i.e., RARL, CARL, DDQN, DADR, and MRHOF, have relatively higher energy consumption under the same conditions, and the energy consumption increases more significantly with the increase of the number of nodes. This is because the GATR algorithm introduces the GAT mechanism, which can combine network topology structure to accurately evaluate local observation states, thereby optimizing routing decisions and reducing unnecessary energy consumption.

Figure 6 further presents the network lifetime performance of the GATR algorithm compared to other comparative algorithms in different network scenarios. The results show that the GATR algorithm always achieves better network lifetime performance in various network scenarios. Specifically,

when $N = 100$, $TR = 120$ ppm/node, the GATR algorithm has increased the network lifetime by approximately 19.4%, 22.7%, 39.9%, 57.3%, and 146.5% compared to DDQN, RARL, DADR, CARL, and MRHOF, respectively. In addition, with the increase of traffic load, although the network lifetime of all algorithms decreases, the GATR algorithm has the smallest decrease, and the GATR algorithm performs better than other algorithms in the network lifetime of different network densities. This is mainly due to the variance sensitive multi-objective reward function design adopted by GATR, which can dynamically balance multidimensional objectives such as residual energy, data transmission reliability, and latency, guiding the model to pay more attention to the network lifetime and stability in the optimization process, thereby extending the network lifetime while ensuring the basic network performance. The GATR algorithm combines the advantages of the GAT mechanism and intelligent reward function, which can significantly improve the energy efficiency and network lifetime while ensuring basic network performance, providing an ideal solution for resource-constrained scenarios such as wireless sensor networks.

5.3.2 Data transmission reliability

Moreover, we evaluate the data transmission reliability of the proposed algorithm through Packet Delivery Rate (PDR) and Buffer Packet Loss Ratio (QLR). PDR can directly reflect the proportion of successfully delivered packets, demonstrating the algorithm's capability to ensure data integrity during transmission. QLR measures the packet loss in the buffer, highlighting the algorithm's effectiveness in handling buffer overflow and preventing packet loss under different network conditions. By integrating PDR and QLR, a comprehensive evaluation of the algorithm's data transmission reliability can be achieved. Specifically, Fig. 7 shows the PDR per-

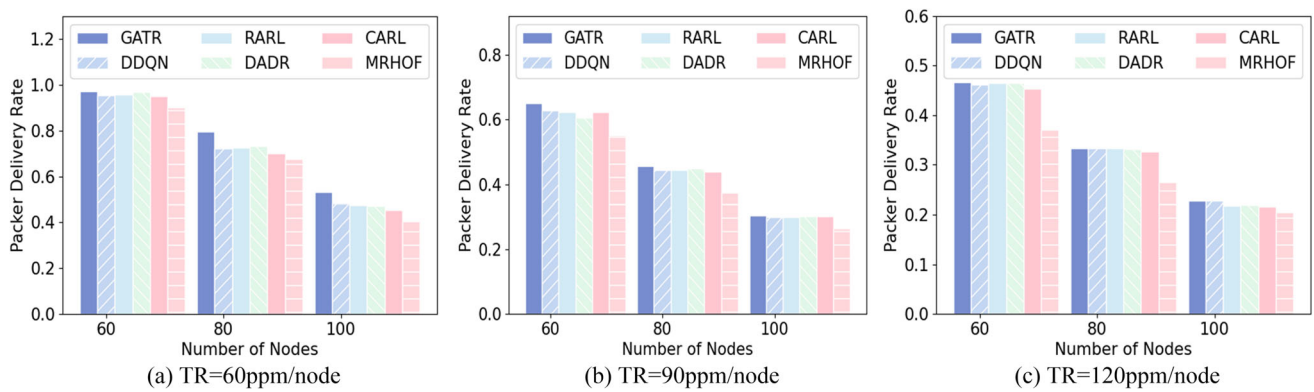


Fig. 7 Comparison of packet delivery rate in different network scenarios

formance of six algorithms under varying network scales and traffic intensities. The results demonstrate that while all algorithms exhibit declining PDR trends with increasing node density and traffic load, GATR consistently maintains superior performance across all scenarios. In light traffic conditions (Fig. 7(a), TR = 60 ppm/node), GATR achieves higher PDR compared to baseline methods while showing the smallest performance degradation with network expansion. This indicates better stability and robustness of GATR. In addition, in moderate-to-heavy traffic scenarios (Fig. 7(b) and (c)), all algorithms demonstrate comparable performance, while GATR retains better scalability and congestion resistance in most cases.

In addition, Fig. 8 further illustrates the average QLR results. Specifically, in light load (Fig. 8(a), TR = 60 ppm/node), all algorithms maintain relatively low QLR. In addition, as the network density increases, the QLR of all algorithms shows an upward trend. This is due to the continuous increase in the number of data packets transmitted simultaneously in the network, which increases the processing burden on routing nodes and links, and makes it easier for queues to overflow beyond capacity limits, thereby increasing packet loss rates. Nevertheless, GATR

always achieves better QLR performance. In Fig. 8(b) and (c), GATR and DDQN substantially outperform other methods in QLR reduction, particularly in high-density networks under heavy loads, where GATR effectively controls packet loss and minimizes transmission failures. The performance gains of GATR in PDR and QLR are attributed to three aspects: 1) The GAT mechanism enables dynamic adjustment of communication weights based on neighbor node states, enhancing topological awareness and routing flexibility; 2) the dual-critic network combined with contrastive learning improves state representation discriminability while mitigating Q-value overestimation; 3) the variance-sensitive term in the reward function facilitates simultaneous optimization of transmission efficiency, energy balance, and load distribution, thereby reducing network hotspots and congestion risks.

5.3.3 Delay

Figure 9 illustrates the end-to-end delay performance comparison among various algorithms under different transmission rates. The experimental results indicate that the MRHOF algorithm achieves the lowest delay across all tested sce-

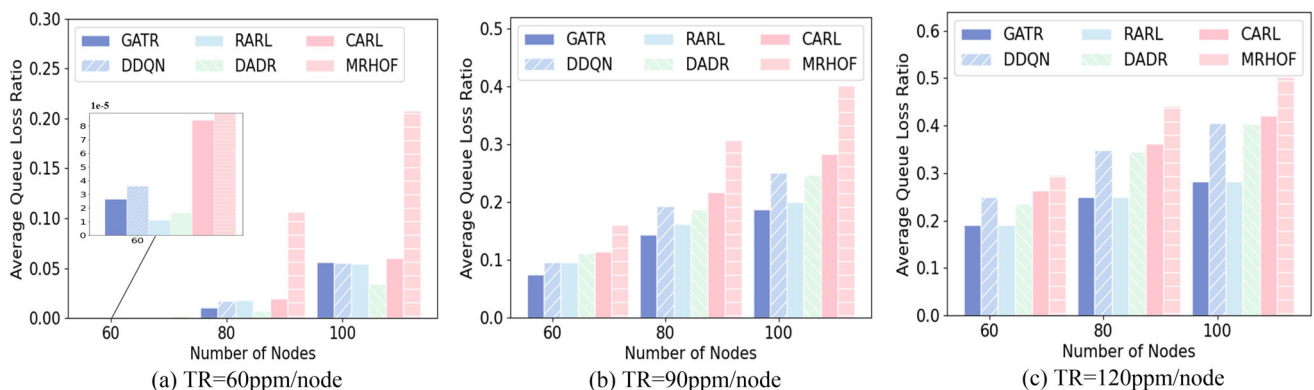


Fig. 8 Comparison of buffer packet loss ratio in different network scenarios

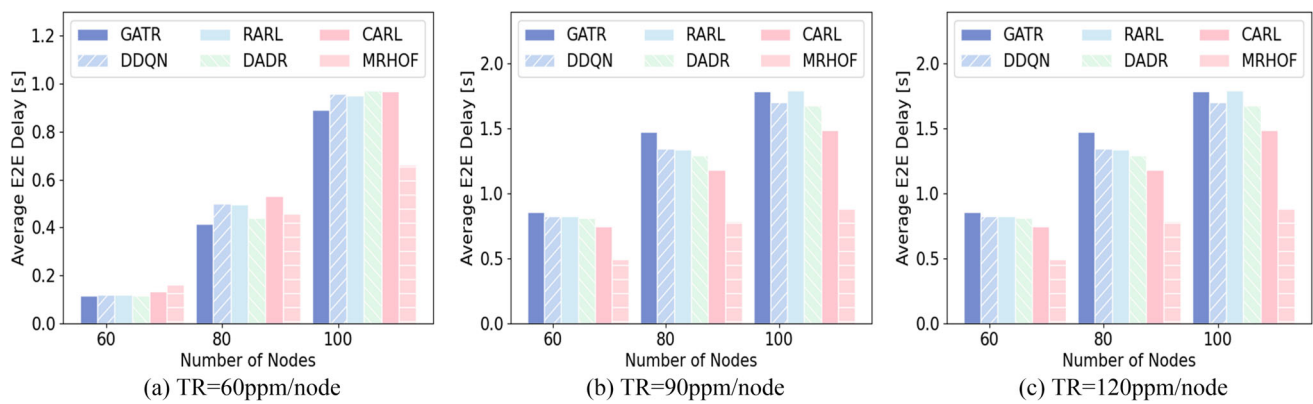


Fig. 9 Comparison of delay in different network scenarios

narios, but this performance advantage comes at the cost of significantly increased energy consumption and reduced transmission reliability. In contrast, the GATR algorithm demonstrates more competitive delay performance in low-load scenarios (as shown in Fig. 9(a)), with only a 0.23s increase in delay compared to MRHOF under the condition of 60 ppm/node when $N = 100$, while achieving a 10.7% reduction in energy consumption and an 31.9% improvement in PDR. This is attributed to its optimized routing decision mechanism and efficient topology awareness capability. However, as the network load increases to 120 ppm/node ($N = 100$), the delay performance of GATR relative to MRHOF increases by 1.2s, mainly due to its multi-objective optimization strategy: it dynamically balances key metrics such as delay, energy consumption, and reliability through a variance-sensitive reward function, prioritizing network stability and energy efficiency as the load increases to avoid over-optimization of a single metric that could lead to an overall performance imbalance. Specifically, under high-load conditions, GATR achieves a 45.8% reduction in energy consumption and a 11.6% improvement in PDR by appropriately

relaxing delay constraints (within an acceptable range). This trade-off design makes GATR more suitable for practical application scenarios that require consideration of multi-dimensional performance indicators.

5.3.4 Comprehensive analysis

Figure 10 is derived from the same simulation results presented in Figs. 5–9, focusing on the representative scenario with 100 nodes under three traffic rates (i.e., 60, 90, and 120 ppm/node). Five performance metrics are considered in this comparison, i.e., PDR, delay, energy consumption, QLR, and network lifetime. These results are first collected from the aforementioned experiments and then normalized to ensure comparability across different dimensions. Specifically, a min–max normalization is applied to each metric across all algorithms, scaling the values to the range [0, 100]. Positive metrics such as PDR and network lifetime are normalized directly, while negative metrics such as delay, energy consumption, and QLR are inverted before normalization so that higher values consistently indicate better perfor-

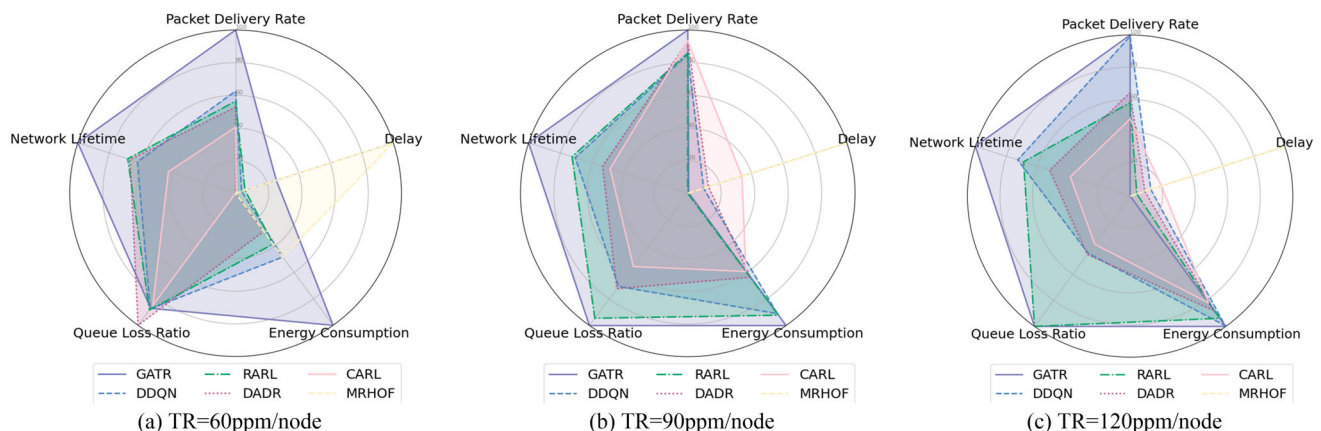


Fig. 10 Comparison of comprehensive performances in different network scenarios. All metrics have been normalized and reversed where necessary, so that higher is always better

Table 2 Performance comparison of GATR vs. w/o GAT, where $TR = 60$ ppm/node

Scenarios	Methods	EC (pJ)	NL	PDR	Delay (s)	QLR
N=60	GATR	1.305	4782	0.971	0.115	2.64×10^{-5}
	w/o GAT	1.329	1881	0.962	0.133	9.030×10^{-4}
N=80	GATR	1.701	4273	0.796	0.415	0.011
	w/o GAT	1.925	1851	0.723	0.509	0.072
N=100	GATR	2.566	4386	0.532	0.890	0.056
	w/o GAT	2.677	2254	0.538	0.953	0.106

Performance metrics include energy consumption (EC), network lifetime (NL), packet delivery ratio (PDR), delay, and queue loss rate (QLR)

mance. Based on these normalized results, the radar chart provides an integrated visualization that highlights the relative strengths and weaknesses of different algorithms under varying network conditions. It can be observed that the proposed GATR algorithm achieves the best results in energy consumption and network lifetime, while maintaining competitive performance in PDR and delay. These results confirm the effectiveness of GATR's multi-objective optimization design, which significantly improves energy efficiency and network sustainability while ensuring network quality.

5.3.5 Impact of GAT on performance

To further highlight the contribution of the GAT, we compare the proposed GATR algorithm with a variant where the GAT modules in the main critic and target critic networks are removed (denoted as w/o GAT). Both variants share identical actor structures and training settings, isolating the effect of GAT. As shown in Table 2, GATR consistently improves energy efficiency and network reliability across all network scenarios ($N = \{60, 80, 100\}$, $TR = 60$ ppm/node). When $N = 60$, GATR reduces energy consumption by 1.8%, extends network lifetime by 154%, and achieves higher PDR with a substantially lower QLR. For $N = 80$, GATR achieves 11.6% energy reduction, 131% longer lifetime, 10.1% PDR improvement, and an 84.7% drop in QLR. In the $N = 100$ setting, energy consumption decreases by 4.2%, network lifetime extends by 95%, and QLR declines by 47%. These results confirm that GAT effectively exploits topological relationships among nodes, enabling more precise state evaluation and robust routing under dynamic, partially observable IoT environments, thereby enhancing critic guidance and yielding significant gains in energy efficiency and reliability.

6 Conclusion

This paper proposes the GATR algorithm to address the critical challenges of routing optimization in IoT networks,

including localized observation limitations, dynamic topology fluctuations, and computational constraints of resource-constrained devices. By integrating the Actor-Critic architecture under the CTDE paradigm, GATR innovates through a topology-aware dual-channel graph modeling mechanism that combines the GAT mechanism for dynamic neighborhood aggregation with edge matrix encoding of physical interference relationships, a three-layer convolutional Actor network architecture enhanced by contrastive learning, and a variance-sensitive multi-objective reward formulation for adaptive metric balancing. Systematic evaluations validate GATR's superiority over RARL, DDQN, DADR, CARL, and MRHOF algorithms across energy efficiency, network lifetime, and data transmission reliability in most network scenarios, demonstrating its effectiveness in balancing conflicting optimization objectives under resource constraints. The algorithm's graph-based framework naturally supports network topologies, while its lightweight design ensures compatibility with low-power devices, and adaptive reward mechanisms enhance robustness against environmental fluctuations. This work presents a practical approach to intelligent IoT routing that supports energy efficiency, reliability, and scalability in communication systems.

Although the GATR algorithm shows superior performance in energy efficiency, network lifetime, and reliability. However, its delay under high-load conditions is slightly inferior compared with some baselines. This is mainly due to the lightweight Actor network, which reduces computational overhead but may limit fine-grained state capture, and the variance-sensitive multi-objective reward, which prioritizes energy and lifetime under heavy traffic. Future work will focus on enhancing state representation through multi-scale or deeper Actor architectures, and adopting hierarchical or adaptive reward mechanisms to further improve delay performance and policy adaptability.

Author Contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Ying Wang, Jianjun Lei, Yuanyuan Li, and Fengjun Shang. The first draft of the manuscript was written by Ying Wang, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding This research is jointly sponsored by Science and Technology Innovation Key R and D Program of Chongqing: CSTB2023TIAD-STX0016, National Natural Science Foundation of China (62276037), and Chongqing talent group Project: cstc2024ycjh-bgzxm0156.

Data Availability The data used to support the findings of this study are available upon request to the corresponding author.

Declarations

Conflicts of Interest The authors declare no conflicts of interest.

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